

DESi ablation — is the gain selection, governance, or just structured context?

Falsification-oriented extension of the ab_evidence A/B harness. This reports the **instrument and the deterministic input-side result**; it is not an empirical verdict on DESi.

Backend status: **UNAVAILABLE_in_this_env**. No API key in this environment, so accuracy & degeneration are **UNAVAILABLE** and **not simulated** (per the harness's no-mock discipline).

Conditions

- A — baseline: the full chat history.
- B — normal DESi: the case's own categorised state slice (id+what per category).
- C — wrong-slice: B's format/budget, but another case's state (cross-domain donor).
- D — status-stripped: B's claim texts with governance metadata removed.
- E — budget-matched status-stripped: D's texts, no metadata, padded with inert filler to B's token budget (~1%).

Input-side characterisation (deterministic, no model)

slice_info_recall = fraction of the case's true ground-truth items actually inside the injected slice. It shows the wrong slice (C) really starves the model of correct information (**0.0**) while B and D carry it (**1.0**); A has the full chat. This describes the **inputs**, not model behaviour.

Core cases

case	condition	input_tokens	slice_info_recall
case1_architecture	A	653	-
case1_architecture	B	736	1.0
case1_architecture	C	613	0.0
case1_architecture	D	444	1.0
case1_architecture	E	743	1.0
case2_research	A	748	-
case2_research	B	655	1.0
case2_research	C	736	0.0
case2_research	D	433	1.0
case2_research	E	662	1.0
case3_debugging	A	697	-
case3_debugging	B	614	1.0
case3_debugging	C	656	0.0
case3_debugging	D	406	1.0
case3_debugging	E	621	1.0
case4	A	4263	-
case4	B	1753	1.0
case4	C	631	0.0
case4	D	1125	1.0
case4	E	1760	1.0

Density sweep (padding grows A; slice constant)

case	condition	input_tokens	slice_info_recall
case6	A	8579	-

case	condition	input_tokens	slice_info_recall
case6	B	3287	1.0
case6	C	631	0.0
case6	D	2057	1.0
case6	E	3294	1.0
case7a_30k	A	26797	-
case7a_30k	B	3287	1.0
case7a_30k	C	673	0.0
case7a_30k	D	2057	1.0
case7a_30k	E	3294	1.0
case7b_60k	A	53196	-
case7b_60k	B	3287	1.0
case7b_60k	C	673	0.0
case7b_60k	D	2057	1.0
case7b_60k	E	3294	1.0

Note: D is ~35–40% smaller than B (that gap IS the governance metadata's footprint). **E** controls for it — D's texts padded with inert filler to B's budget (within ~1%) — so only **B>E** (not merely B>D) would be evidence for the metadata. Runner: temp 0, fixed seed, 3 reps/case.

Degeneration metrics (first-class, deterministic, unit-tested)

- **loop_trap** — the answer repeats near-identical units (stuck in a loop)
- **contradiction_persistence** — a known open conflict restated as if settled
- **invalid_claim_reuse** — the answer restates a claim from a known-invalid pool
- **bad_framing_nonrecovery** — share of the answer traceable to a wrong injected slice
- **coherence_without_continuity** — fluent output that preserves little of the true state

What would weaken vs strengthen the DESi claim

Weaken: $C \approx B$ on accuracy (gain is generic structured context); $D \approx B$ (metadata decorative); $A \geq B$ except at extreme density (long-context artefact).

Strengthen: C collapses / high bad_framing_nonrecovery (selection is load-bearing); **B > E** (budget-matched) on conflict/decision typing and degeneration (metadata governs, not token count); B's edge and lower degeneration grow with density (robustness under pressure).

Limitations (stated plainly)

No backend here (instrument only, no verdict). Tiny sample (4 core + 3 density) — no significance. D token-budget confound now controlled by E. Paraphrase-blind evaluator (relative comparison only). Wrong-slice budget only approximate. Negative results are primary; no promotional claims.

Reproduce: `python ab_evidence/ablation_run.py` (deterministic) · set `OPENROUTER_API_KEY` then re-run for the full table · `pytest tests/ab_ablation -q`.